

Paolo Di Francesco, PhD

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Senior Solutions Architect at AWS with 15+ years in tech spanning cloud architecture, MLOps, AI/GenAI, cybersecurity, and telecommunications research. Published author, public speaker, and external university lecturer.

PROFESSIONAL EXPERIENCE

Amazon Web Services (AWS)

Vienna, Austria

Senior Solutions Architect

Jul 2022 – Present

- Contributed code to open-source projects including [MLflow](#) (27K+ stars) and the [Amazon SageMaker Python SDK](#) (2K+ stars).
- Engaged in design discussions with the AWS service team during the development of managed MLflow on Amazon SageMaker AI, contributing customer-driven feedback grounded in my upstream MLflow contributions and field experience running MLflow on AWS.
- Re-imagined the developer portal experience with agentic AI: built and presented an AI-orchestrated infrastructure bootstrap based on [Strands Agents SOPs](#), deploying a web application across 3 AWS accounts (~150 resources, 6 Terraform modules) in under one hour with human approval gates — replacing a multi-week manual runbook process.
- Published 7 technical blog posts on the [AWS Machine Learning Blog](#) (54K+ views).
- Developed public workshops used by AWS field teams worldwide and delivered to thousands of customers:
 - [Amazon SageMaker Immersion Day](#) (GitHub)
 - [MLOps: from idea to production in six steps](#) (GitHub)
- Facilitated multi-day hackathons and innovation events for enterprise customers, with projects going to production.
- Delivered presentations to audiences of 200+ at AWS Summits, Cloud Days, meetups, and university conferences in English and Italian.
- Mentored and onboarded fellow Solutions Architects; contributed to building the local AI/ML community in Austria.

Solutions Architect

2020 – Jul 2022

- Helped customers across startups, SMBs, and enterprises adopt AWS services and design scalable cloud architectures.
- Led cloud migration and adoption programs, including multi-account landing zone setups and cost optimization.
- Delivered immersion days and architecture reviews across industries including manufacturing, media, energy, and technology.

RadarServices GmbH

Vienna, Austria

Software Engineer & Architect

Oct 2017 – 2020

- Designed and implemented a service-oriented architecture for a cloud-based IT security monitoring product. Led the transformation of a large monolith into a modular, scalable system.
- Defined business processes with enterprise customers and implemented solutions according to their needs.
- Designed and documented REST APIs. Resolved critical performance bottlenecks to meet strict SLAs. Mentored junior developers.

Development Team Leader

Jan 2017 – Oct 2017

- Led a development team ensuring consistent delivery. Managed hiring and stakeholder collaboration.

Software Developer

Mar 2016 – Dec 2016

- Test-driven development with peer review.

Trinity College Dublin

Dublin, Ireland

Postdoctoral Researcher

Sep 2015 – Mar 2016

- Contributed to the [Fed4FIRE](#) EU FP7 project, developing frontend/backend for remote access to the TCD wireless testbed. Designed a web-based resource management and monitoring system.

Teaching Assistant

Sep 2012 – May 2015

- Taught Java programming and Fundamentals of Electrotechnics (filtering, modulation, oscilloscope usage).

Research Assistant

Mar 2010 – Dec 2010

- Developed a split-architecture for carrier-sensing based MAC protocols on SDR platforms. Collected experimental data to analyse protocol performance in multi-hop wireless networks.

O2 Ireland & Trinity College Dublin

Dublin, Ireland

Data Analyst (university-enterprise collaboration)

Sep 2013 – Sep 2014

- Collected and maintained large datasets of anonymized call detail records. Analysed spatio-temporal dynamics in cellular traffic demand to inform network planning research.

TEACHING

TU Wien

Vienna, Austria

External Lecturer – AI and Generative AI module

Mar 2023 – Present

- Lecturing for the [Project Course on Cloud Computing \(192.035\)](#), covering foundation models, prompt engineering, RAG, and building AI-powered applications on the cloud. Multiple cohorts of graduates.

EDUCATION

Ph.D. in Electronic & Electrical Engineering

Trinity College Dublin, 2011 – 2016

Thesis: “On the Effects of Resource Sharing on Mobile Network Deployment Decisions”

Focus: Algorithm Design, Mathematical Modeling, Linear Optimization, Game Theory, Data Analysis

M.Sc in Telecommunications Engineering

University of Bologna, 2008 – 2011

B.Sc in Telecommunications Engineering

University of Bologna, 2004 – 2008

TECHNICAL SKILLS

Cloud & AI/ML: Amazon SageMaker AI, Amazon Bedrock, AWS Lambda, Amazon API Gateway, AWS CDK, AWS CloudFormation, Amazon ECS, MLflow, Docker, MLOps

Programming: Python, Java, JavaScript, Ruby, C/C++, MATLAB

Frameworks & Tools: Ruby on Rails, CDK, Git, LaTeX, GNU Radio, Gurobi

Databases: PostgreSQL, MySQL

LANGUAGES

Italian (native) English (fluent) German (basic)

AWARDS & CERTIFICATIONS

- **Invent & Simplify Award** (2022) – recognized by the AWS Solutions Architecture community for contributions to MLOps and open-source
- Member of the winning team at the **IEEE DySPAN 5G Spectrum Challenge 2015**

- AWS Certified Machine Learning – Specialty (2021) AWS Certified Solutions Architect – Professional (2021)
- AWS Certified Solutions Architect – Associate (2020) AWS Certified Developer – Associate (2020)
- AWS Certified SysOps Administrator – Associate (2020)

PUBLIC SPEAKING

- “From Runbooks to Production: AI-Orchestrated Infrastructure at Enterprise Scale” – AWS AI SDLC Workshop, Vienna, Jun 2026
- “FMOPs/LLMOPs – Managing the Generative AI Lifecycle on AWS” – University of Applied Sciences Burgenland, Mar 2025
- “Accelerate FM pre-training on Amazon SageMaker HyperPod” – AWS Summit Zurich, Sep 2024
- “LLMOPs and FMOPs – Evaluating and Operating Foundation Models” – AWS Builder Stream, Vienna, Mar 2024
- “Productionize ML workloads using Amazon SageMaker MLOps” – AWS Cloud Day Zurich, Sep 2023
- “Foundation Model Hosting and RAG on AWS” – Vienna Data Science Tools Meetup, Aug 2023
- “AWS DeepRacer Workshop” – AWS Summit Milan, Jun 2023
- “MLOps on AWS” (Italian track) – AWS Innovate AI/ML Edition, Feb 2023
- “MLOps – Automazione per il Machine Learning” – AWS Italy Podcast, Oct 2022
- “MLflow on AWS and SageMaker Integration” – AWS Vienna Meetup (organizer/host), Sep 2022

SELECTED PUBLICATIONS

Journals

- **P. Di Francesco**, F. Malandrino, and L.A. DaSilva, “Assembling and Using a Cellular Dataset for Mobile Network Analysis and Planning,” *IEEE Trans. on Big Data*, 4(4), 2018.
- **P. Di Francesco**, F. Malandrino, and L.A. DaSilva, “Sensitivity Analysis on Service-Driven Network Planning,” *IEEE/ACM Trans. on Networking*, 25(3), 2017.
- **P. Di Francesco**, F. Malandrino, T. Forde, and L.A. DaSilva, “A Sharing and Competition Aware Framework for Cellular Network Evolution Planning,” *IEEE Trans. on Cognitive Communications and Networking*, 1(4), 2015.
- **P. Di Francesco**, S. McGettrick, U.K. Anyanwu, A.B. MacKenzie, and L.A. DaSilva, “A Split MAC Approach for SDR Platforms,” *IEEE Trans. on Computers*, 64(4), 2015.

Conferences

- J. Kibilda, **P. Di Francesco**, F. Malandrino, and L.A. DaSilva, “Infrastructure and Spectrum Sharing Trade-offs in Mobile Networks,” *IEEE DySPAN*, 2015.
- **P. Di Francesco**, F. Malandrino, and L.A. DaSilva, “Mobile Network Sharing Between Operators: A Demand Trace-Driven Study,” *ACM SIGCOMM CSWS*, 2014.
- Y. Xiao, Y. Chau, **P. Di Francesco**, and L.A. DaSilva, “Dynamic Spectrum Scheduling for Carrier Aggregation: A Game Theoretic Approach,” *IEEE ICC*, 2013.
- A. Puschmann, **P. Di Francesco**, M.A. Kalil, L.A. DaSilva, and A. Mitschele-Thiel, “Enhancing the Performance of Random Access MAC Protocols for Low-cost SDRs,” *ACM WiNTECH*, 2013.
- **P. Di Francesco**, S. McGettrick, U.K. Anyanwu, J.C. O’Sullivan, A.B. MacKenzie, and L.A. DaSilva, “A Split Architecture for Random Access MAC for SDR Platforms,” *CROWNCOM*, 2013.
- L.A. DaSilva, J. Kibilda, **P. Di Francesco**, T.K. Forde, and L.E. Doyle, “Customized Services over Virtual Wireless Networks: The Path towards Networks without Borders,” *Future Network and MobileSummit*, 2013.
- J.C. O’Sullivan, **P. Di Francesco**, U.K. Anyanwu, L.A. DaSilva, and A.B. MacKenzie, “Multi-hop MAC Implementations for Affordable SDR Hardware,” *IEEE DySPAN*, 2011.